

Temperature & Air Quality

Atlantic Provinces, Canada

Yashaswi Tiwari | May 2026



TEMPERATURE

Changes in Temperature and Air pollutants across Atlantic provinces

Is Atlantic, P...

^ ATLANTIC

■ NB

■ NL

■ NS

■ PEI

∨ ☐ REST OF CA...

Year

☐ 1995

☐ 1996

☐ 1997

☐ 1998

☐ 1999

☐ 2000

☐ 2001

☐ 2002

☐ 2003

☐ 2004

☐ 2005

Pollutant

CO

NO2

O3

PM10

PM2.5

SO2

5.60

Avg Monthly Temp °C

6.80

Avg Monthly Prediction °C

0.31

Trend (°C /Decade)

Good

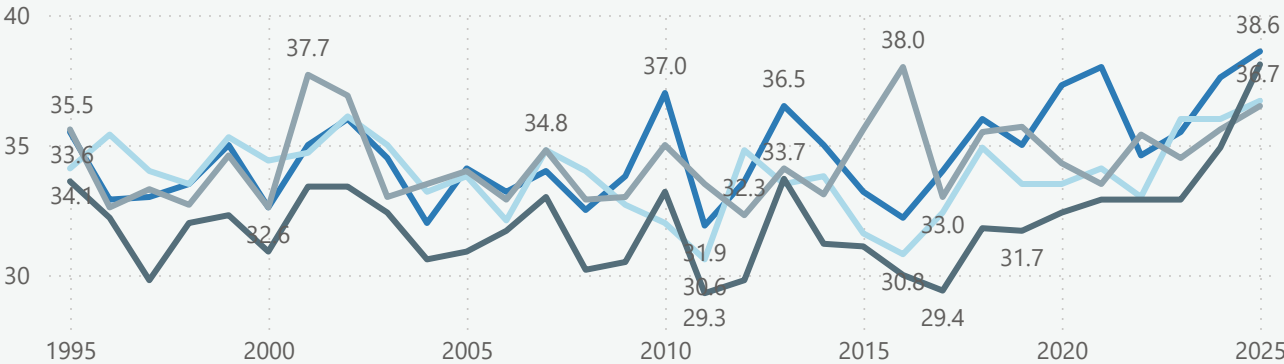
Avg AQI

166

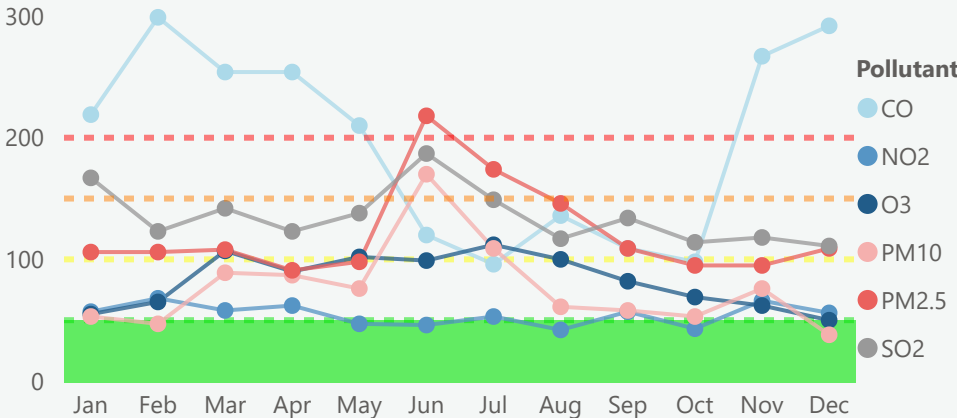
Unhealthy Days

Max Temp (°C)

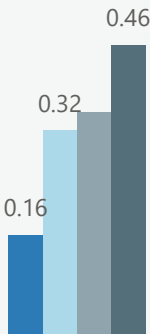
Province_Code ● NB ● NL ● NS ● PEI



Worst Pollutant AQI



Trend (°C / decade) GHG Temp Correlation



PEI

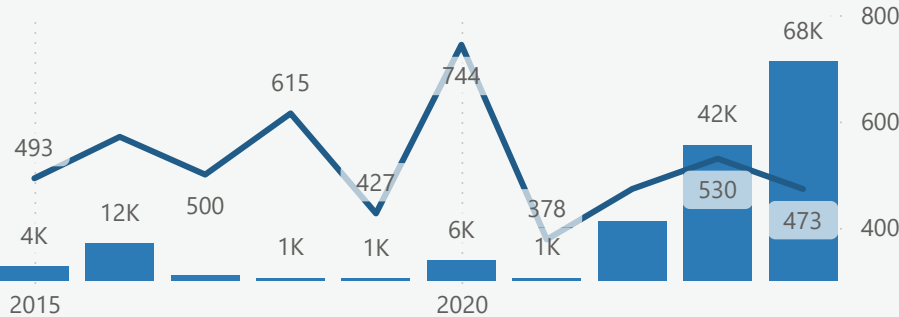
0.19

Avg Correlation

Correlation

Pollutant	Mean Temp (°C)
PM2.5	0.16
O3	0.15
PM10	0.08
CO	0.04
SO2	-0.01
NO2	-0.01

● Total Area Burned ● Total Fires



5.60
Avg Monthly Temp(°C)

-0.08
Temp Monthly Anomal...

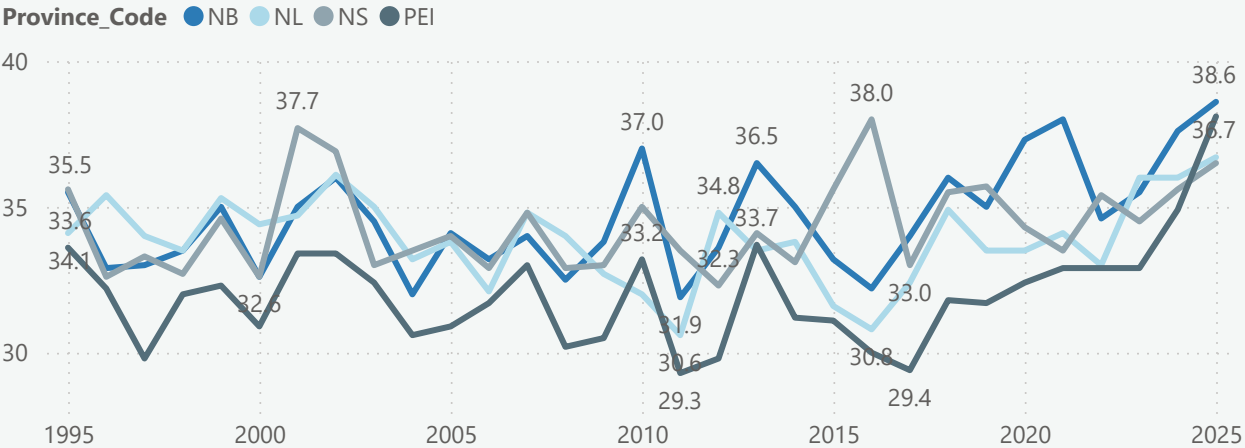
0.31
Trend (°C /Decade)

76.77
Avg Monthly Heatwaves

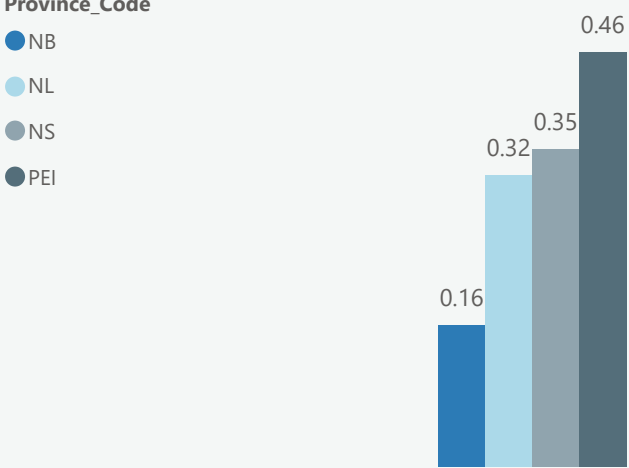
6.80
Average Prediction(°C)

- Provin...
☐ NB
☐ NL
☐ NS
☐ PEI
- Season
☐ Fall
☐ Spring
☐ Summer
☐ Winter

Max Temp (°C)



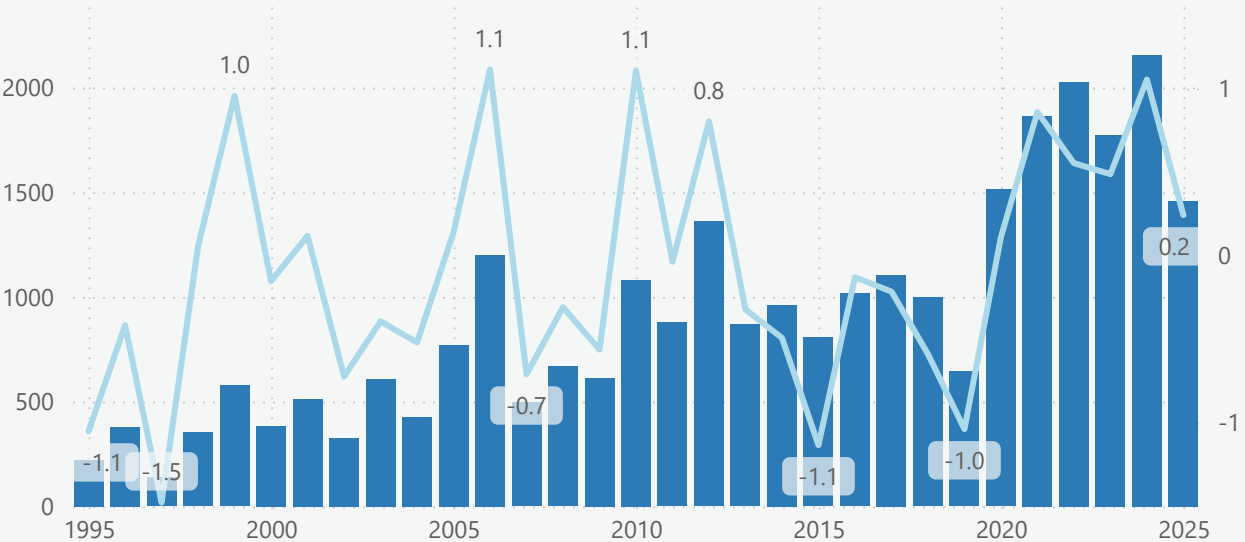
Trend (°C / decade)



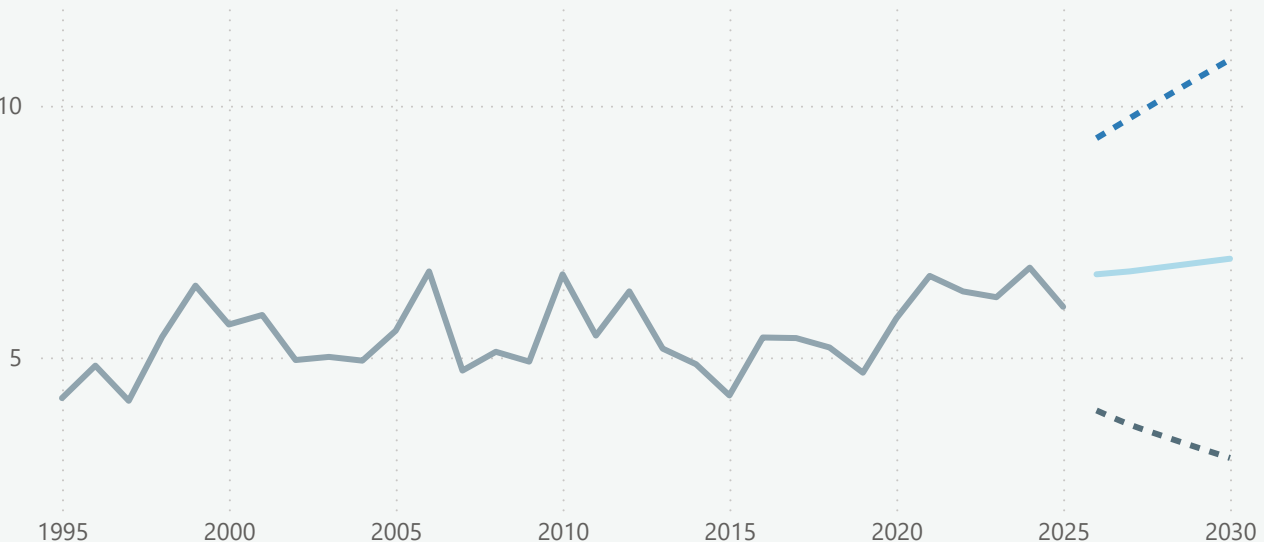
Model Comparison

OLS: MAE: 7.4763
RMSE: 8.3924
ARIMA: MAE: 1.5652
RMSE: 1.9148
SARIMA: MAE: 0.9765
RMSE: 1.2408

Heatwave Count



Upper Bound(°C)



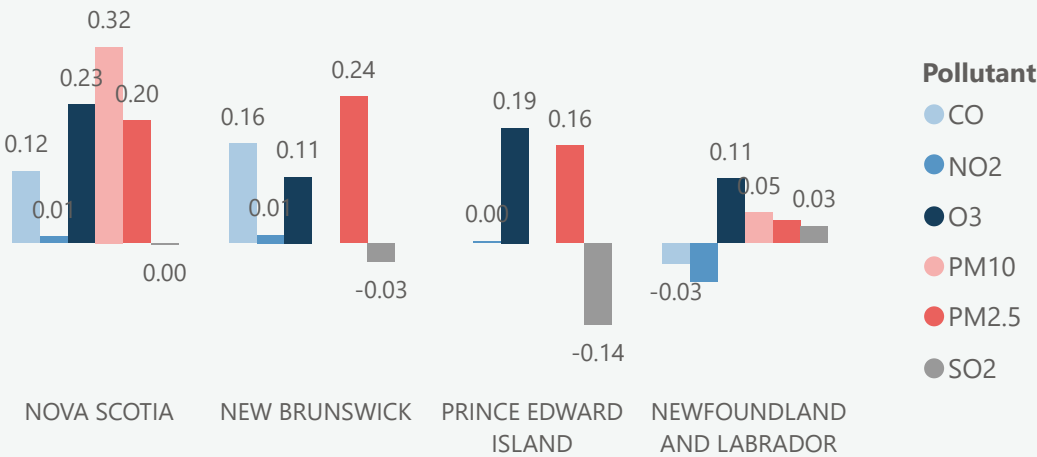
CORRELATION OF POLLUTANT CONCENTRATIONS & GREEN HOUSE GAS EMISSIONS WITH TEMPERATURE

- Season
- Fall
 - Spring
 - Summer
 - Winter

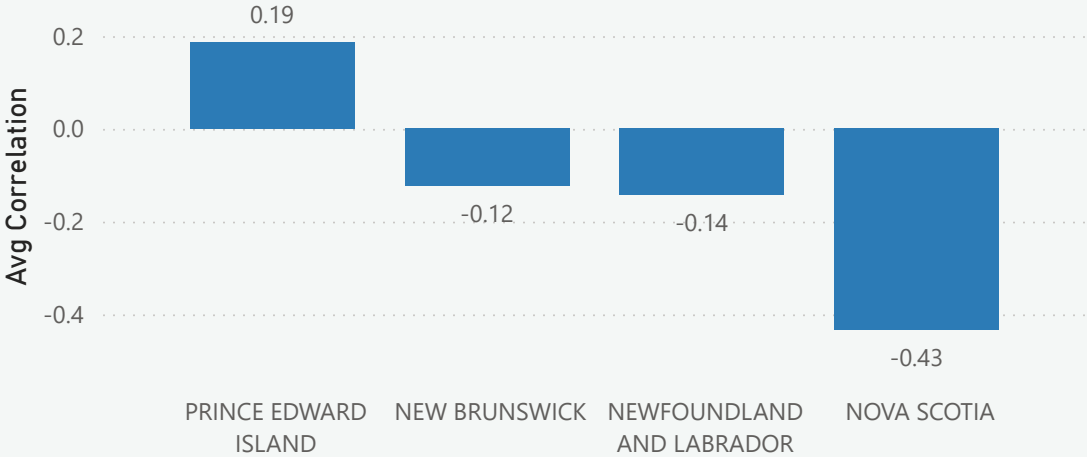
- Province co...
- NB
 - NL
 - NS
 - PEI

- Province
- NB
 - NL
 - NS
 - PEI

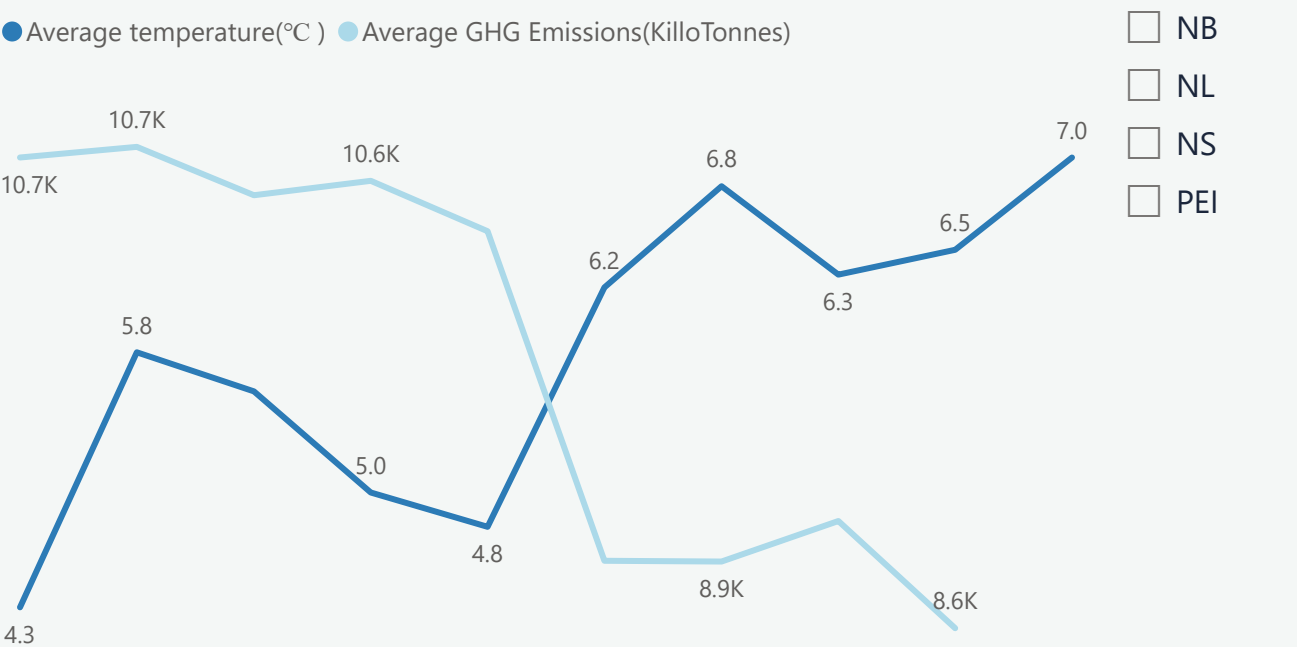
Pollutant Temperature Correlation



GHG Temperature Correlation



Pollutant	Mean Temp (°C)	Spd of Max Gust (km/h)	Total Rain (mm)	Total Snow (cm)
CO	0.04	-0.20	-0.08	-0.01
NO2	-0.01	-0.19	-0.03	0.00
O3	0.15	0.19	0.05	-0.01
PM10	0.08	0.05	-0.11	-0.09
PM2.5	0.16	-0.04	-0.12	-0.07
SO2	-0.01	-0.03	-0.07	-0.02



Good

Station AQI

34.7

Avg Station AQI

O3

Top Pollutant

PM2.5

Second Pollutant

0.16%

% Unhealthy Days

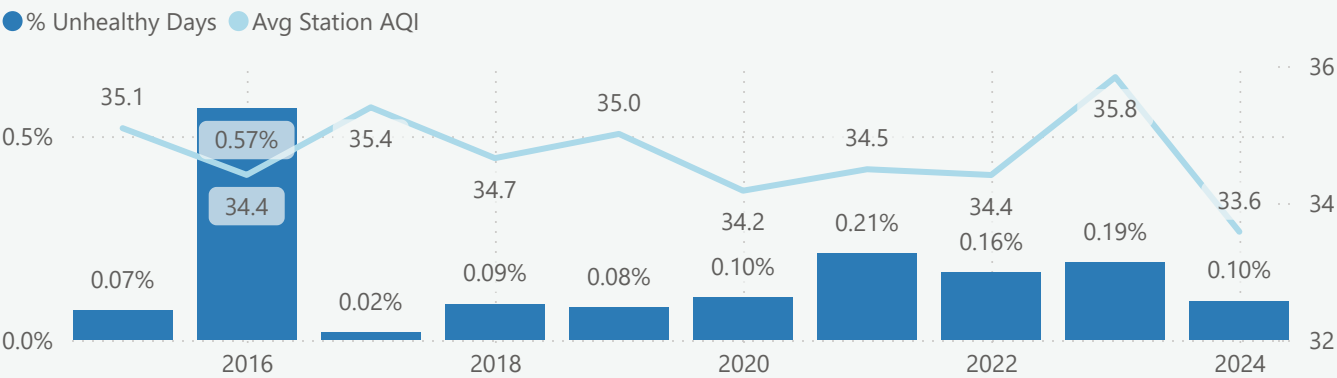
299

Worst Pollutant AQI

Overall AQI is GOOD ✓ (Avg: 34.7), with 0.2% unhealthy days. Trend is improving ↗, however watch for worst AQI level which reached 299 on 25 Feb 2016.

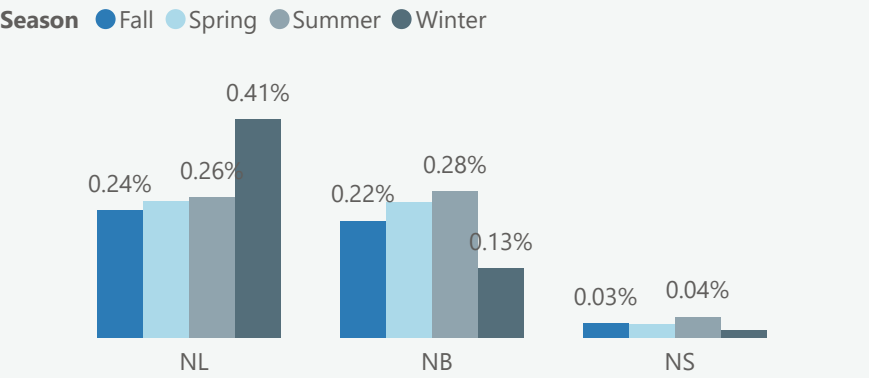
AQI Insight

% Unhealthy Days and Avg Station AQI



Province	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
+ NEW BRUNSWICK	21.1	16.2	20.6	19.6	19.9	19.3	19.5	19.4	19.4	17.5
+ NEWFOUNDLAND AND LABRADOR	15.0	17.0	15.7	14.6	14.0	13.8	13.9	14.2	13.0	14.5
- NOVA SCOTIA										
CO	2.7	2.1	1.9	2.2	1.8	1.7	1.3	1.8	2.1	1.8
NO2	7.8	5.8	7.2	6.6	7.0	6.6	5.5	5.9	5.7	5.8
O3	30.3	29.1	30.6	31.8	31.4	30.9	31.4	31.7	32.0	31.5
PM10								10.3		
PM2.5	29.9	28.5	31.3	29.2	27.5	26.8	27.4	26.2	27.4	21.6
SO2	3.1	2.5	2.8	3.2	2.6	2.4	2.2	2.7	2.1	1.6
+ PRINCE EDWARD ISLAND	17.8	20.4	18.5	15.9	18.9	18.2	17.2	16.6	16.6	16.6

% Unhealthy Days



Avg Station AQI



Clear all slicers

Is Atlantic, Provin...

^

ATLANTIC

NB

NL

NS

PEI

Year

2014

2015

2016

2017

2018

2019

2020

2021

2022

2023

2024

Season

Fall

Spring

Summer

Winter

Station_Type

AQI Station

166

Unhealthy Days

CO, O3, SO2, PM2.5, PM10

Unhealthy Pollutants (>100 AQI)

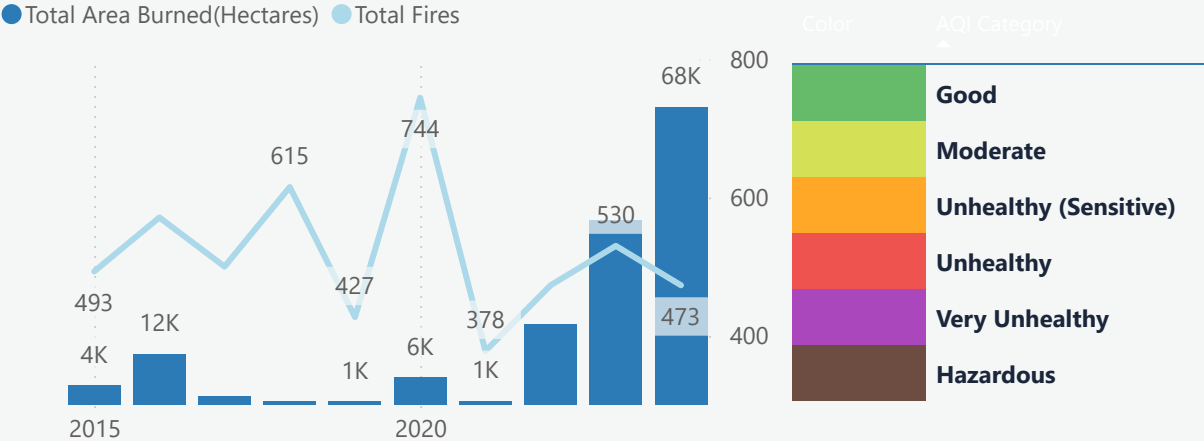
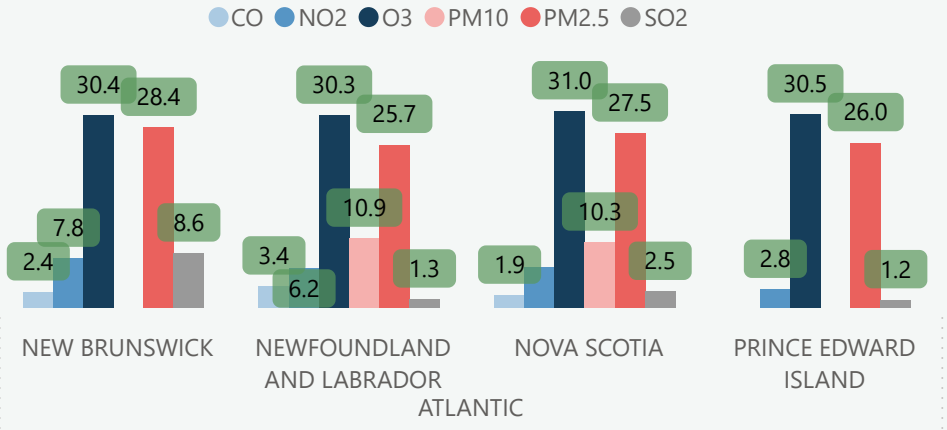
Is Atlantic, Provin...

- ^ ATLANTIC
- NB
- NL
- NS
- PFI

Year

- 2015
- 2016
- 2017
- 2018
- 2019
- 2020
- 2021
- 2022
- 2023
- 2024

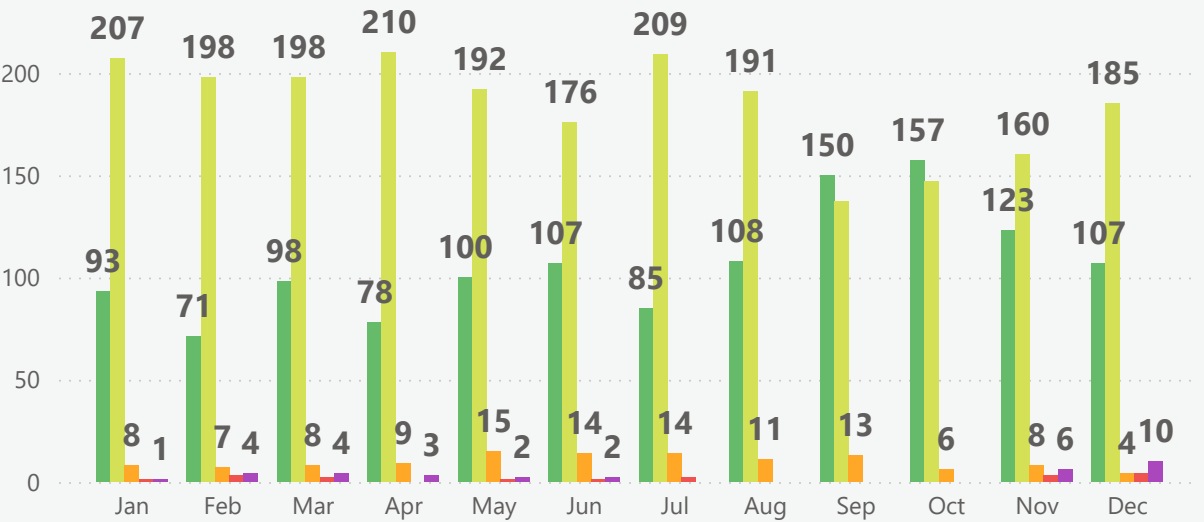
Avg Pollutant AQI



Worst Day: Pollutant AQI

Province_Code	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
NL	219	299	254	254	210	218	174	146	109	107	267	292
NB	167	123	142	123	138	187	149	117	134	114	118	111
NS	80	101	97	105	98	98	112	114	102	105	78	73
PEI	79	76	80	70	96	72	76	80	82	69	71	66

Days in AQI Category



Pollutant

CO

NO2

O3

PM10

PM2.5

SO2

Station_Type

A.

C.



Good

Station AQI

38.5

Avg Station AQI

O3

Top Pollutant

PM2.5

Second Pollutant

1.19%

% Unhealthy Days

300

Worst Pollutant AQI

Overall AQI is GOOD ✔ (Avg: 38.5), with 1.2% unhealthy days. Trend is improving ↗, however watch for worst AQI level which reached 300 on 03 Sep 2023.

AQI Insight

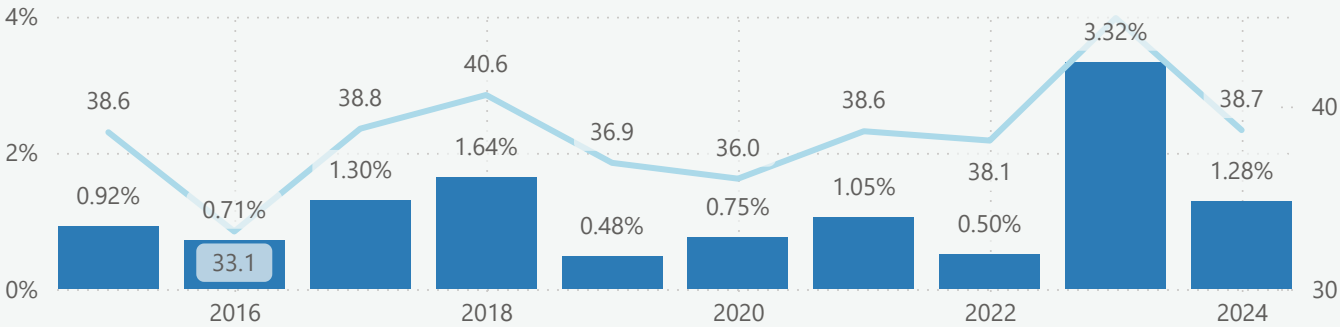
Is Atlantic, Provin...

✓ ☐ ATLANTIC

✓ ☒ REST OF C...

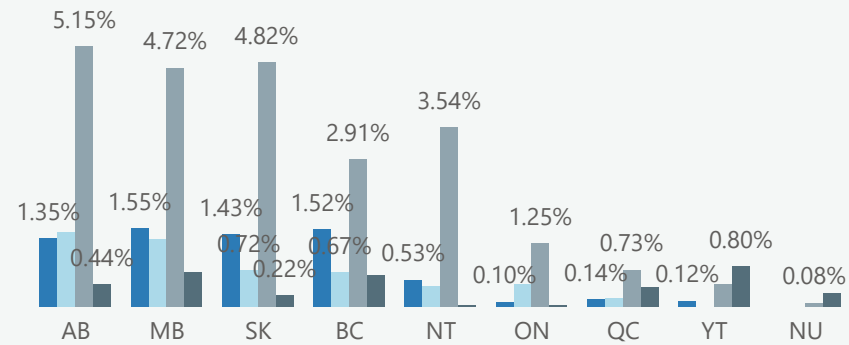
% Unhealthy Days and Avg Station AQI

● % Unhealthy Days ● Avg Station AQI



% Unhealthy Days

Season ● Fall ● Spring ● Summer ● Winter



Province	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
ALBERTA	18.0	15.5	18.8	20.3	18.0	17.1	18.6	18.0	21.7	19.4
BRITISH COLUMBIA	18.8	15.5	19.7	19.8	18.1	17.7	18.3	18.3	18.6	16.8
MANITOBA	18.8	17.9	21.2	22.5	17.7	18.8	22.3	17.3	21.3	20.8
NORTHWEST TERRITORIES	14.3	11.0	14.0	13.7	14.5	12.5	14.5	15.6	19.7	19.5
NUNAVUT	25.9	20.1	20.7	20.2	21.2	29.0	29.7	27.9	30.1	29.2
ONTARIO	27.3	25.6	25.0	25.4	24.6	23.8	25.1	24.5	25.5	21.5
QUEBEC	26.0	22.2	25.2	25.2	24.2	23.4	24.4	24.5	21.9	24.0
SASKATCHEWAN	18.5	14.7	19.4	20.8	19.4	18.5	21.1	19.6	23.4	21.2
YUKON	23.3	20.8	22.0	20.6	24.6	23.7	24.6	22.8	21.9	22.1

Average Station AQI



Year

☐ 2015

☐ 2016

☐ 2017

☐ 2018

☐ 2019

☐ 2020

☐ 2021

☐ 2022

☐ 2023

☐ 2024

Season

☐ Fall

☐ Spring

☐ Summer

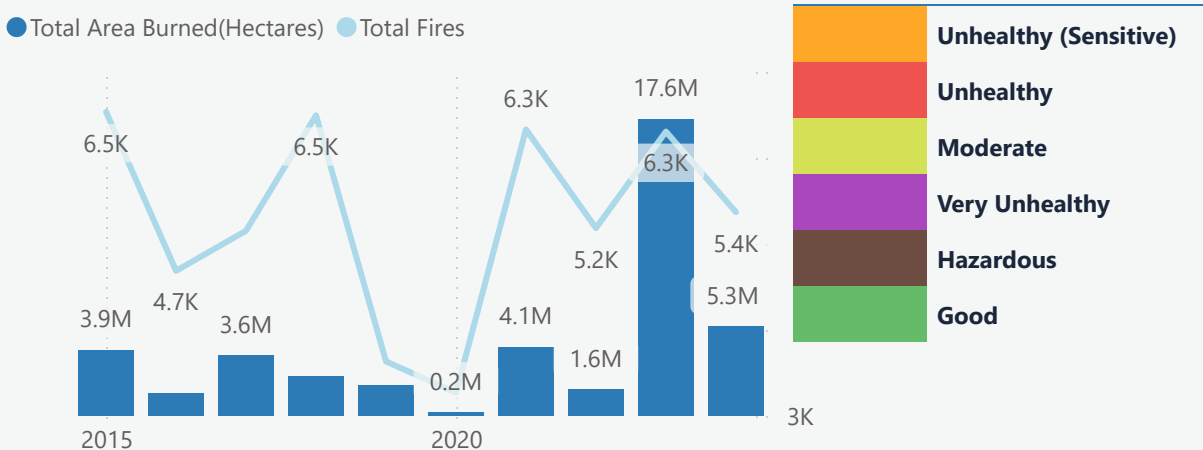
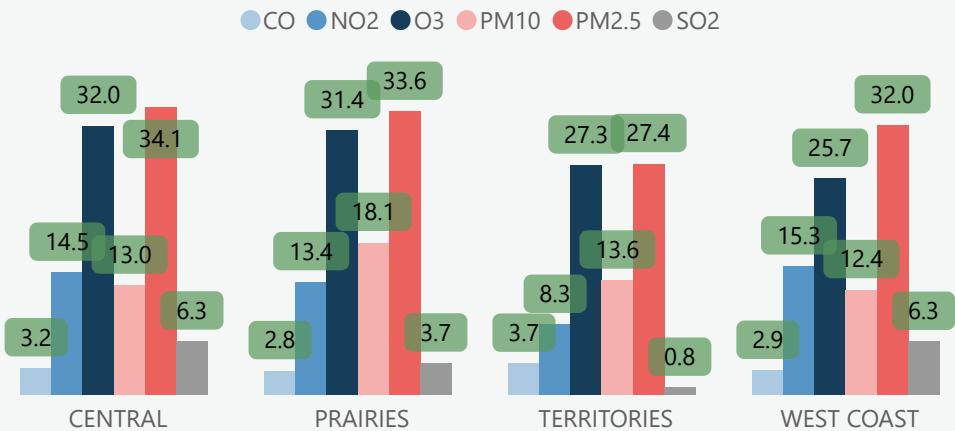
☐ Winter

Station_Type

☒ AQI Station



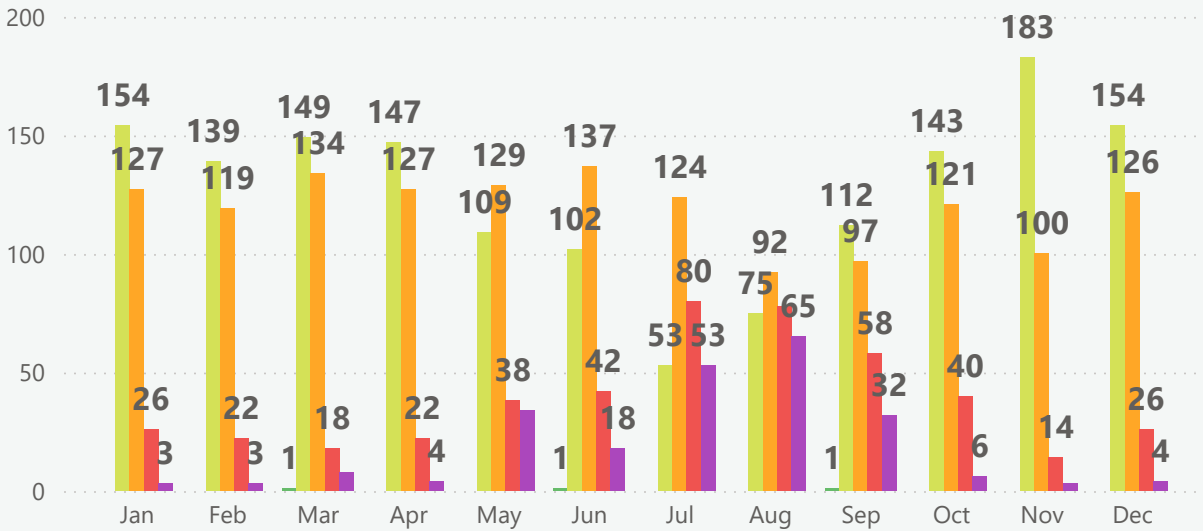
Avg Pollutant AQI



Worst Day: Pollutant AQI

Province_Code	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
⊕ AB	172	194	283	208	300	296	297	297	300	272	233	178
⊕ BC	215	197	254	193	288	210	280	295	297	283	189	242
⊕ NT	79	105	156	167	277	214	295	297	277	157	70	94
⊕ MB	197	255	254	213	268	245	289	259	288	185	205	171
⊕ SK	116	147	126	163	286	284	279	239	279	224	139	114
⊕ ON	106	124	173	128	175	279	166	169	135	163	138	137
⊕ QC	157	239	139	121	150	273	159	152	140	153	156	188
⊕ NU	260	260	79	55	61	44	54	260	62	88	79	103
⊕ YT	136	113	81	64	71	105	151	90	59	115	80	122

Days in AQI Category



Pollutant

- co
- NO2
- O3
- PM10
- PM2.5
- SO2

Station_Type

AQI St...

Data Sources & References:

Climate, Temperature & Precipitation

Environment and Climate Change Canada (ECCC) – Daily Climate Observations

Station-level daily climate data (temperature, precipitation, wind, snow) in CSV format, accessed by province, Climate ID, and year.

<https://dd.weather.gc.ca/today/climate/observations/>

Technical Documentation by ECCC: https://climate.weather.gc.ca/doc/Technical_Documentation.pdf

ECCC – Historical Climate Data Portal

Official portal for historical station records, climate summaries, and validated climate normals.

<https://climate.weather.gc.ca/>

Canadian Centre for Climate Services – Climate Data Access

Centralized access to historical climate datasets, metadata, and documentation used for baseline and anomaly analysis.

<https://www.canada.ca/en/environment-climate-change/services/climate-change/canadian-centre-climate-services/display-download/climate-data.html>

Air Quality Index (AQI)

Environment and Climate Change Canada – Air Quality Health Index (AQHI)

National AQI datasets and health-based index documentation.

<https://www.canada.ca/en/environment-climate-change/services/air-quality-health-index.html>

Government of Canada Open Data – Air Quality Datasets

Open station-level air quality measurements used for AQI trend and exceedance analysis.

https://search.open.canada.ca/opendata/?search_text=air+quality

<https://data-donnees.az.ec.gc.ca/data/air/monitor/national-air-pollution-surveillance-naps-program?lang=en>

Greenhouse Gas (GHG) Emissions

Canadian Environmental Sustainability Indicators (CESI) – Greenhouse Gas Emissions

Annual provincial and national GHG emissions data by economic sector.

<https://www.canada.ca/en/environment-climate-change/services/environmental-indicators/greenhouse-gas-emissions.html>

Government of Canada Open Data – National GHG Inventory

Machine-readable datasets supporting province-level and sector-level emissions analysis.

(<https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3810009701&cubeTimeFrame.startYear=2013&cubeTimeFrame.endYear=2023&referencePeriods=20130101%2C20230101>)

Wildfire data

Wildfire data- Statistics are shown for all reported fires

The Canadian National Fire Database (CNFDB) is a collection of forest fire data from various sources; these data include fire locations (point data) and fire perimeters (polygon data) as provided by Canadian fire management agencies (provinces, territories, and Parks Canada).

The National Burned Area Composite (NBAC) is a GIS database and system that calculates the area of forest burned on a national scale for each year since 1972. The Canadian National Fire Database (CNFDB) SRC:

NFDB_point_20250519 - <https://cwfis.cfs.nrcan.gc.ca/datamart/metadata/nfdbpnt> <https://www.canada.ca/en/environment-climate-change/services/air-quality-health-index.html>